

KYAMBOGO UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF BIOMEDICAL AND MECHATRONICS
ENGINEERING
BACHELOR'S OF ENGINEERING IN MECHATRONICS
AND BIOMEDICAL ENGINEERING

COMPUTING II
END OF SEMESTER 2 COURSEWORK AND PRESENTATION

TITLE:
Designing a Prepaid Water Credit System

BY: GROUP C (EVENING)

EPUWAT ELLY BENON	25/U/BIE/18668/PE
AMALI MAUREEN LYDIA	25/U/BIE/05286/PE
MUWONGE RONALD PAUL	25/U/BIE/01401/PE
DEMBE MARK	25/U/BIE/01366/PE
GIZAMBA JOEL	25/U/BIE/01370/PE

Lecturer: Mr. OKELLO WAYNE STEVEN
Date: May 26, 2026

Contents

1	Introduction	2
2	System Constants and Conditions	2
3	Declaration of Functions	2
4	The Use of Structures	3
5	Mathematical Business Logic	3
5.1	Calculation of Gross Amount	3
5.2	Calculating Number of Credit Units	3
6	General operation of the program	3
7	Main Requirements of the Program	4
8	Conclusion	5

1 Introduction

The system is a program designed to manage utility consumption in this case water consumption for either domestic or commercial meters. The system is called “HAVEN PREPAID WATER SYSTEM”. It enters the amount of money paid by a customer or user and calculates the number of units for the amount paid.

2 System Constants and Conditions

The program required authentication and some fixed values for calculations. These values of constants are listed below:

- **System Access PIN:** 4545
- **Fixed Maintenance Fee Deducted:** 500.0 UGX
- **Domestic Unit Rate:** 150.0 UGX per Unit
- **Commercial Unit Rate:** 250.0 UGX per Unit
- **Leakage Alert Safety Limit:** 50.0 Units per day

3 Declaration of Functions

The dynamic prototype configurations include:

- `int authenticate ();` – Validates user access by requesting for the system security PIN.
- `void addHousehold ();` – Allocates storage and activates a new customer record profile.
- `void searchHousehold ();` – Looks through the existing system registry to locate a targeted household profile.
- `void updateHousehold ();` – Modifies parameters or utility variables within an existing record.
- `void deleteHousehold ();` – Permanently removes a consumer record from the active memory of the program.
- `void buyWaterCredit ();` – Evaluates transaction inputs, applies deductions, and prints tokenized units.
- `void displaySummary ();` – Compiles and prints comprehensive billing and usage registries.

4 The Use of Structures

Structures are basically used to store data. Unlike arrays structures store data in different data types.

```
typedef struct {  
    int id;  
    char headName[50];  
    char type[15];  
    float balanceUnits;  
    float dailyUsageUnits;  
} Household;
```

5 Mathematical Business Logic

The program works in a way that when a user or household owner makes a deposit(money), M, it deducts a maintenance fee, F_m of 500UGX and the balance(grossAmount), M_{net} is used to calculate the units bought by dividing by the dynamicRate (for either domestic or commercial meter). The mathematical evaluation is modeled below:

5.1 Calculation of Gross Amount

Given a gross money M, the processing grossAmount M_{net} is computed after a fixed maintenance fee (F_m) extraction: If M less than F_m , the operational routine flags an immediate structural evaluation error and terminates execution.

$$M_{net} = M - F_m$$

5.2 Calculating Number of Credit Units

The final number of credit units (U) are scaled dynamically based on a string mapping logical condition checking account categories: Where: Dynamicrate = 250.0 UGX, if Type = " Commercial"

150.0 UGX, if Type = " Domestic"

$$\Delta U = \frac{M_{net}}{\text{DynamicRate}}$$

6 General operation of the program

- i. **All-in-One Variable:** We created a custom struct to tie a user's ID, Name, Account Type, Balance, and Usage together.
- ii. **Account Type Flag:** A simple number identifies whether the user is Domestic or Commercial.
- iii. **Login Gatekeeper:** A basic login function checks the user before showing the main menu.

- iv. **Master PIN:** The system is protected by a simple access pin.
- v. **Pass or Fail:** If the password is right, it returns 1 to unlock the app; otherwise, it closes down.
- vi. **Continuous Loop:** The program keeps running inside a loop so the admin can do multiple tasks without restarting.
- vii. **Menu Routing:** A switch-case block takes the user's choice (1 to 7) and jumps directly to that specific task.
- viii. **Wrong Input Guard:** If someone types a wrong number or letter, the program catches it and asks them to try again.
- ix. **Registering Users:** Before adding a new household, the code checks if the list is full or if the ID already exists.
- x. **Searching by ID:** A basic for loop goes through the list one by one to find the matching ID.
- xi. **Not Found Warning:** If the loop finishes and finds nothing, it safely tells the admin the record doesn't exist.
- xii. **Editing Info:** Once an ID is matched, the admin can directly type over the old name or update the daily water usage.
- xiii. **Deleting a Record:** To remove someone, the program locates their spot and slides all the remaining names up by one index to fill the gap.
- xiv. **Updating the Count:** The system automatically lowers the total user count so it doesn't leave blank spaces in the memory.

7 Main Requirements of the Program

- i. To receive household information and store it using structures.
- ii. To be able to edit, update and delete the household information.
- iii. To calculate the number of credit units depending on how much a user has input.
- iv. The program should print a receipt after the calculation of the credit units showing the units purchased and the balance of units.
- v. It should be able to print the summary of records of households stored in the system showing all their details including credit units and balance.
- vi. The program should be able to detect leakage and low credit balance and display an appropriate error or warning message.

8 Conclusion

The application of code and the use of c programming provided an efficient tool to

Automate the calculation of water utility consumption. By replacing manual billing methods, the software minimizes human error and significantly speeds up data processing. The system also prints and displays precise and summarized data from transactions like receipts and summary tables of stored information. The program also has room for further

Modification and improvement into better versions and could also be modified for other

Operations and systems in the future besides billing systems.